

Nikita Kirnosov

Principal Engineer / Engineering Manager

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Skills

Design High Load Systems, Recommender Systems, ML Platforms Design, Distributed Systems, Data ETL, Performance Optimization, Machine Learning

Languages C/C++, Python, Java, Go, SQL

Experience

2021 Apr – **Staff Software Engineer / Engineering Manager, Google - Search.**

Present Created and managed a unified Search Growth Infrastructure team of 10 engineers, changed user-content matching approach, server/device work split, and brought the most recent innovations to a stack. In this role I have established and led cross-org collaboration of 87 engineers from Google Research, Core Infrastructure, and Android.

- Created a team powering the full cycle ML ecosystem (indexing, matching, training, serving, calibration):
 - defined development process and reduced incident rate 25x from the previously used infrastructure solutions,
 - introduced machine learned request parameter tuning and saved 83% of compute while improving latency by 147%,
- Augmented heuristic-based content-user matching rules with ML based content retrieval, improving user engagement by 64%.
- Introduced on-device ML to a stack, improving notification triggering quality by 27%.
- Streamlined real-time ML and encoder-decoder architecture recommendation models training and experimentation (details & impact NDA).
- Integrated large language models (LLMs) with the stack (details & impact NDA).

2018 Apr – **Senior Software Engineer, Google - Search.**

2021 Apr Envisioned and lead the team implementing ML training and serving ecosystem for Search's News and Content Discovery.

- Designed and lead the implementation of unified and standardized ML infrastructure for Discover, Notifications, Suggest and user modeling teams featuring:
 - signal retrieval and rich inference serving platform (C++),
 - zero training/serving skew near real time training data generation system, reducing the feature engineering work by 70% across the org (Go, C++),
 - reinforcement learning support (in collaboration with DeepMind),
 - dynamic triggering rate optimization based on recent and long-term user activity pattern (C++).
- Scaled an existing inference serving infrastructure to serve 24x more inferences using the same resources.
- Restructured Discover Search training process to support training on TPU to achieve 40x speedup (Python).
- Launched a flexible accelerator resources acquisition and scheduling solution to improve GPU and TPU utilization by 140% (in collaboration with DeepMind).
- Created user opt-out detection system proactively stopping bad experiments (C++).

2016 Apr – **Software Engineer, Google - YouTube.**

2018 Apr Built storage abstraction microservices framework with no-downtime migration support (C++) and improved YT Public Data API performance, usability, and security (Python).

- Designed and implemented storage management framework allowing the user to create a full set of standard RESTful APIs for a defined resource in several hours regardless of the storage type (C++).
- Migrated video metadata migration between Vitess and Spanner with no downtime, data loss, or client latency increase (C++).
- Analyzed data access patterns and added implicitly needed methods to the API (Python).
- Simplified fraud detection by applying Fourier transform to visualize client requests in frequency domain (Python).

2015 **Fellow, Insight Data Science.**

Sep–Oct Built [CityGravity.info](#), a trip schedule optimization app to see the greatest number of attractions.

2010 Aug – **Graduate Researcher, University of Arizona, Tucson, AZ.**

2016 Jan Developed Fortran code modeling the molecular system total wave function as an optimal combination of explicitly correlated Gaussian functions. Generalized the approach to be applied to exotic atoms and molecules and extended it to excited states. The results are published in over a dozen peer-reviewed papers and presented on multiple physics and supercomputing conferences.

2010 **Intern, Joint Institute of Nuclear Research, Dubna, Russia.**

Jan–May Contributed to muon detector for CMS software development by assessing signal processing code and vectorizing it.

Education

2010–2015 **Ph.D., University of Arizona, Tucson, AZ – Physics.**

2005–2010 **B.S. (Hons.), Saratov State University, Saratov, Russia – Physics.**